

TRACING DECODER ARTIFACTS FOR COMPACT SYNTHETIC SPEECH SCREENING

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ABSTRACT

Recent advances in speech synthesis and voice cloning have increased the need for reliable synthetic-speech detection, yet high-accuracy detectors increasingly rely on large pretrained models that are costly to invoke on every recording. Rather than replacing such detectors, we investigate a compact front-end screen that processes all inputs cheaply and forwards only suspicious recordings for more expensive analysis. To enable lightweight screening without a large learned encoder, we exploit spectral traces introduced by speech-generation operations. We analyze how learned upsampling and inverse short-time Fourier transform synthesis can produce predictable spectral artifacts and measure their presence directly in generated waveforms. Because the strength of these artifacts varies across generators, we combine decoder-guided spectral measurements with complementary descriptors of short-time spectral shape and temporal variation in a compact gradient-boosted tree. Across seven speech generators and two human-speech sources, the proposed screen achieves an equal error rate of 0.021% with an estimated model storage of 151 KiB. When used as the first stage of a simulated cascade with a 1.15-billion-parameter detector, it reduces estimated detection energy by 84.4% while operating at a 0.050% synthetic-speech miss rate, demonstrating the potential of decoder-guided acoustic evidence for low-cost front-end screening.

Index Terms— synthetic speech detection, decoder artifacts, lightweight screening, spectro-temporal analysis

1. INTRODUCTION

Recent advances in text-to-speech synthesis and voice cloning have enabled high-quality AI voices for applications such as voice assistants, customer service, online meetings, and social media. At the same time, increasingly realistic synthetic speech creates new opportunities for impersonation and fraud, making reliable synthetic-speech detection important for trustworthy voice communication. High-performing detectors increasingly rely on large pretrained speech encoders to capture rich acoustic representations. For example, Speech DF Arena, a recent benchmark that evaluates open-source and proprietary speech deepfake detectors across diverse datasets and attack conditions [1], shows that many leading systems contain hundreds of millions to billions of parameters. Although such models provide strong detection performance, applying them to every incoming recording can impose substantial computational and energy costs, particularly in high-volume or always-on applications.

Rather than replacing these high-capacity detectors, we investigate a different deployment strategy: using a compact model as a front-end screen. The screen processes every recording locally and forwards only suspicious inputs to a more expensive backend detector. Such a design can reduce unnecessary large-model inference and, when screening is performed locally, can also reduce the amount of audio that must be transmitted for remote analysis. Importantly, the objective of such a screen differs from that of a standalone detector. A

useful front end should remain inexpensive enough to run on every input while maintaining a sufficiently low synthetic-speech miss rate, so that most benign recordings can bypass expensive downstream processing.

This requirement raises a fundamental representation question: *what acoustic evidence can support reliable screening without relying on a large learned encoder?* We explore traces introduced by the speech-generation process itself. Prior work shows that vocoder-resynthesized speech can provide useful evidence for detecting speech generated by complete text-to-speech and voice-conversion systems [2]. In particular, waveform decoders commonly use learned upsampling or inverse short-time Fourier transform (iSTFT) synthesis, whose periodic operations can leave structured spectral traces in generated audio [3]. These traces are attractive for lightweight screening because they can be measured directly from the waveform and linked to identifiable operations in the synthesis pipeline.

Decoder artifacts alone, however, provide incomplete evidence. Their frequencies and prominence depend on decoder architecture, sampling rates, learned filters, and subsequent waveform processing. We therefore combine mechanism-guided artifact measurements with complementary acoustic descriptors. Specifically, we analyze decoder upsampling and iSTFT synthesis to identify frequencies at which narrow spectral peaks may occur, and measure both their local prominence and persistent spectral residuals in the output waveform [4]. We further characterize short-time spectral shape using linear-frequency cepstral coefficients (LFCCs) and summarize their temporal changes through local statistics and modulation spectra, following prior work on temporal variation in synthetic-speech detection [5]. These complementary measurements capture both mechanism-specific spectral traces and broader spectro-temporal differences, and are fused using a compact gradient-boosted tree.

We evaluate the resulting screen on speech from seven generators and two human speech sources. It achieves an equal error rate of 0.021% with an estimated model storage of 151 KiB, approximately 78% smaller than the smallest neural comparator evaluated in our study. We further examine generalization to unseen generators and analyze the model as the first stage of a detection cascade. In a simulated deployment with a 1.15-billion-parameter backend detector, operating the screen at a 0.050% synthetic-speech miss rate reduces estimated detection energy by 84.4%. Together, these results show that decoder-guided acoustic measurements can provide a compact first-stage screening mechanism that avoids unnecessary large-model inference while retaining interpretable connections to the speech generation process.

2. RELATED WORK

Recent speech deepfake datasets broaden coverage to more speakers, diverse acoustic conditions, and adversarial attacks [6]. Large-scale multilingual collections also cover text-to-speech, voice conversion, and neural vocoder outputs, enabling evaluation across languages and synthesis methods [7]. Evaluations across multiple datasets

reveal substantial performance variation across sources and recording conditions, showing that strong results on one benchmark do not ensure cross-domain reliability [1].

Self-supervised learning based detectors use speech encoders to provide representations for high-accuracy deepfake detection [8]. These encoders learn from large unlabeled speech corpora, and their representations are used by classifiers trained on labeled detection data. Lightweight detectors pursue lower computational and storage costs through compact networks or explicit acoustic features. Waveform architectures use spectro-temporal graph attention or lightweight convolutions [9, 10], while recent cepstral studies examine MFCCs, LFCCs, and temporal differences for seen-source and cross-source detection [11]. Closely related artifact-based methods learn lightweight detectors from frequency-domain fingerprints [12] or extract residual fingerprints from differences between audio and its filtered version for detection and model attribution [13].

3. ARTIFACT-GUIDED DETECTION

Our detector combines decoder-guided spectral evidence with complementary spectro-temporal descriptors. We first analyze decoder upsampling and iSTFT synthesis to identify frequencies at which generation artifacts may occur and quantify their evidence in the final waveform. We then characterize short-time spectral shape and its temporal variation using cepstral features. The resulting feature groups are fused using a compact gradient-boosted tree.

3.1. Decoder-Guided Spectral Cues

3.1.1. Artifact Origins and Frequencies

Decoder Mechanisms. Many speech generators synthesize waveforms from low-rate acoustic representations, such as mel spectrograms or codec embeddings. Learned upsampling increases temporal resolution, while filters and nonlinear activations shape the signal. These operations can also introduce periodic components whose frequencies depend on decoder sampling rates. Some decoders predict waveform samples directly, as in HiFi-GAN [14], whereas others predict spectral coefficients and reconstruct the waveform using an inverse short-time Fourier transform (iSTFT). We examine how these operations can produce structured spectral peaks in the output waveform. Biases and nonlinear activations, such as Leaky ReLU and Snake, can introduce nonzero temporal means in decoder features. Such a mean corresponds to a zero-frequency, or DC, component that subsequent upsampling can convert into tonal artifacts [3].

Transposed-Convolution Artifacts. A transposed convolution with stride s is equivalent to inserting $s - 1$ zeros between input samples and convolving the expanded sequence with a learned kernel h . For a single input–output channel pair, the response to a constant input component of amplitude μ , away from boundaries and before adding the output bias, is $y_\mu[sm + r] = \mu G_r$, where $G_r = \sum_q h[r + qs]$ for $0 \leq r < s$, with the sum taken over valid kernel indices. Because different output phases receive different subsets of kernel weights, unequal phase sums produce a period- s response when $\mu \neq 0$. This is the one-dimensional checkerboard mechanism expressed through the DC gains of the polyphase filters [15].

In the frequency domain, zero insertion creates spectral replicas separated by the input sampling rate f_{in} , while convolution scales them according to the learned kernel response. A period- s component at output sampling rate $f_{out} = s f_{in}$ can therefore contain components at $f_k = k f_{in}$ for positive integers k within the output Nyquist band. Their amplitudes depend on the feature means, learned filters, and contributions across channels. Later upsampling stages can further replicate or attenuate these components [3].

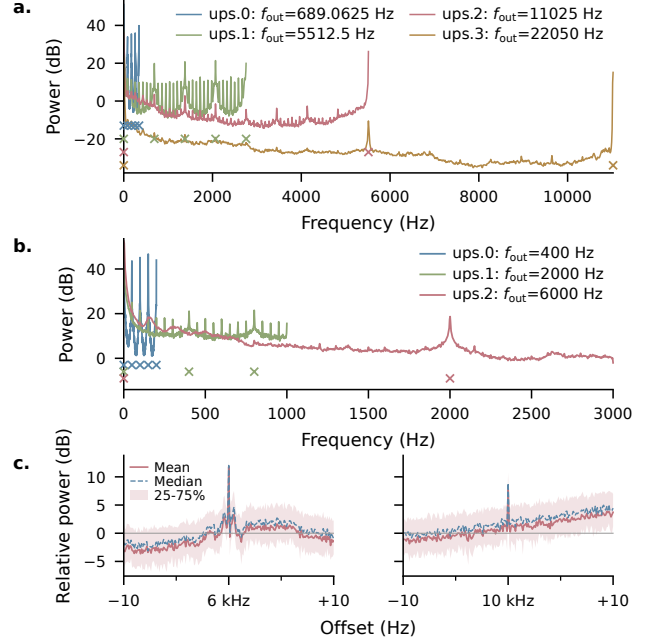


Fig. 1. Upsampling-layer spectra of (a) the OpenVoice V2 tone-color converter and (b) the CosyVoice2 HiFT decoder; crosses mark predicted DC-replica frequencies. (c) Spectral peaks relative to the local background at 6 and 10 kHz across 200 CosyVoice2 recordings.

For example, OpenVoice uses a waveform decoder with output sampling rate $F_s = 22050$ Hz and upsampling strides (8, 8, 2, 2) [16]. For L layers with strides $s_1, \dots, s_L$, the input sampling rate of layer j is $R_{j-1} = F_s / \prod_{\ell=j}^L s_\ell$. The third OpenVoice upsampling layer therefore receives features at 5512.5 Hz, giving a possible DC-replica frequency at 5512.5 Hz. Figure 1(a) shows the spectra of the four upsampling-layer outputs from a single forward pass.

iSTFT Artifacts. For decoders that predict spectral coefficients, artifact locations also depend on waveform reconstruction. In CosyVoice2, the final upsampling layer produces features at a sampling rate of 6000 Hz and exhibits a peak at 2000 Hz in Fig. 1(b). The decoder predicts magnitude and phase to form complex spectral coefficients and reconstructs 24-kHz audio using a Hann-windowed iSTFT with $N = 16$ and hop $H = 4$ [17].

The spectral coefficients can be decomposed into their temporal mean and a zero-mean remainder. The mean repeats the same synthesis frame every H samples; away from boundaries, normalized overlap-add therefore produces an H -periodic contribution with possible spectral components at integer multiples of $F_s/H = 6000$ Hz. Periodic variation retained in the zero-mean component can produce additional replicas around multiples of F_s/H . For example, if the 2000-Hz variation persists in the coefficients, one possible component occurs at $2(F_s/H) - 2000 = 10000$ Hz.

Waveform Evidence. These predicted locations are also visible in generated audio. Figure 1(c) shows narrow spectral peaks near both 6 and 10 kHz in the mean and median spectra of a separate collection of 200 CosyVoice2 recordings, consistent with the decoder analysis. These waveform measurements use a 10-s rectangular-window FFT at 24 kHz (0.1-Hz bins), with the waveform mean removed and median log power at offsets $4 < |\Delta f| < 40$ Hz subtracted.

3.1.2. Waveform Measurements

We measure decoder artifacts from two complementary perspectives: a broadband representation of persistent spectral peaks and decoder-

guided measurements at predicted artifact frequencies.

Broadband Spectral Residuals. To capture persistent peaks throughout the analysis band without assuming their exact locations, we measure their prominence above the surrounding spectral background. For each 16-kHz input, we compute a Hann-windowed STFT and average the log-power spectra across frames. Over 1–7.95 kHz, we apply the background-estimation and normalization procedure of Afchar et al. [4]: a background estimated from local spectral minima is subtracted, and positive residuals are clipped and normalized per recording. We retain the normalized residuals at their original frequency bins, preserving both the locations and relative strengths of persistent peaks. Figure 2(a)–(b) illustrates the background subtraction and resulting residual structure.

Decoder-Guided Frequencies. We next derive candidate artifact locations from a range of neural vocoder and codec decoder configurations, including HiFi-GAN [14], UnivNet [18], and EnCodec [19]. Applying the relations in Section 3.1.1 and merging locations that fall into the same FFT bin yields a fixed set of 23 candidate frequencies within 1–7.95 kHz. Their actual amplitudes depend on the learned filters and subsequent processing, so these frequencies specify where evidence is measured rather than assuming that every generator exhibits a strong peak.

Local Peak Prominence. At each predicted frequency, we measure peak prominence from the mean log-power spectrum before background subtraction, clipping, or normalization. Following the local-background comparison in Fig. 1(c), each frequency is mapped to its nearest FFT bin. Peak prominence is defined as the maximum log power across that bin and its two neighbors minus the median log power of background bins more than 4 Hz and less than 40 Hz away. The neighborhood accommodates FFT-grid mismatch, while the median estimates the local background.

We additionally compare each candidate peak with nearby spectral fluctuations. For each predicted location, we subtract the median prominence at nearby reference centers from the prominence at the candidate frequency. We use six reference centers on each side, 8–40 Hz from the mapped frequency. Background bins and reference centers exclude locations within three FFT bins of registered analysis frequencies and selected replica-grid positions. We retain both measurements separately for every candidate frequency, complementing the broadband spectral residuals with location-specific decoder-guided evidence.

3.2. Spectro-Temporal Features

The decoder-guided measurements above emphasize persistent spectral peaks and are largely based on temporal averaging. Because such artifacts can vary in strength across generators, we complement them with short-time spectral shape and its temporal dynamics. Using 20-ms Hann windows with a 10-ms hop, we compute log energies from 70 linearly spaced triangular filters over 0–8 kHz and apply a discrete cosine transform, retaining the first 40 linear-frequency cepstral coefficients (LFCCs) [20]. Each coefficient forms a temporal trajectory used by two complementary feature branches.

Distributional Statistics. The first branch summarizes the static LFCC trajectories and their first- and second-order temporal differences. We summarize each trajectory using its median, 10th and 90th percentiles, and the interdecile range (the 90th percentile minus the 10th percentile), capturing its typical level and spread while reducing sensitivity to extreme values. These statistics characterize spectral-shape distributions together with local temporal variation.

Temporal Modulation. The second branch characterizes how rapidly spectral shape changes over time, following modulation-based synthetic-speech detection [5]. For each of the 40 static LFCC

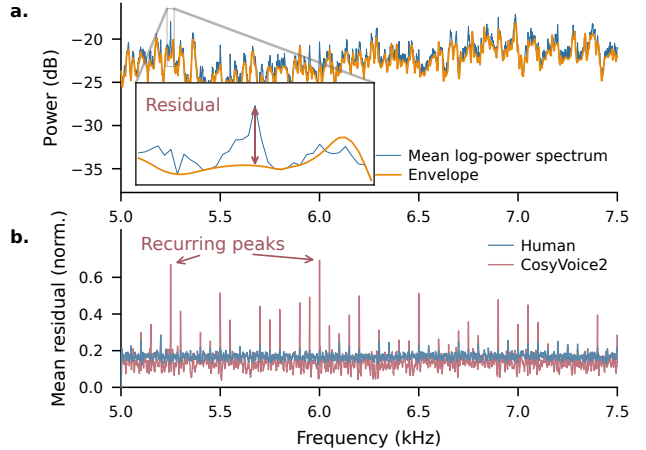


Fig. 2. (a) Background subtraction at 5.25 kHz in a CosyVoice2 recording; the envelope is the background curve estimated from local spectral minima. (b) Mean normalized spectral residuals from 200 LibriSeVoc (Human) and 200 CosyVoice2 (TTS) recordings.

trajectories, we subtract the temporal mean and compute its Fourier magnitude spectrum along the frame axis. With a frame rate of 100 Hz, we average magnitudes within eight logarithmically spaced bands spanning 1–50 Hz and apply logarithmic compression. These modulation features capture the strength of spectral-shape fluctuations at different temporal rates and complement the distributional statistics above.

3.3. Feature Fusion & Classification

We concatenate four feature groups: broadband spectral residuals, decoder-guided peak-prominence measurements, LFCC distributional statistics, and temporal modulation features. The combined representation is classified using a histogram-based gradient-boosted tree with 1000 boosting iterations, at most 15 leaves per tree, a learning rate of 0.1, and at least 20 samples per leaf. Training uses class- and source-balanced weights. We compare feature configurations on validation data and reserve the test set for final evaluation.

4. EXPERIMENTS

4.1. Evaluation Protocols

We evaluate the detector under three complementary settings. The *seen-source* evaluation measures discrimination when training and test recordings come from the same speech sources but share no recordings. The *cross-source* evaluation tests transfer to previously unseen human and synthetic-speech sources without retraining. Finally, leave-one-generator-out (LOGO) evaluation isolates generator dependence by withholding each synthetic-speech generator from training in turn.

Seen-Source Evaluation. Synthetic speech comes from seven SpeechFake BD subsets [7]: Tortoise, MeloTTS, CosyVoice, OpenVoice, Fish-Speech, StyleTTS2, and ParlerTTS. Human speech comes from the bona fide subset of ASvspoof 5 [6] and the ground-truth subset of LibriSeVoc [21]. We split 100,000 development recordings into 80,002 for training and 19,998 for validation, each approximately class-balanced. For our detector, the validation set is further divided into 10,000 tuning and 9,998 calibration recordings. The held-out test set contains 86,234 recordings, equally divided between human and synthetic speech. The training, validation, and test sets share no recordings. Human recordings are drawn 80% from ASvspoof 5 and 20% from LibriSeVoc.

Table 1. Model storage and detection performance: seen-source EER (%) and accuracy (%) at frozen seen-source thresholds on unseen human speech and four additional TTS systems.

System	Size	Seen-Source	Cross-Source				
		EER	Libri [22]	Chatt. [23]	Cosy2 [17]	Qwen3 [24]	Vibe [25]
RawGAT-ST [26]	1.68 MiB	0.023	86.1	100.0	100.0	100.0	100.0
AASIST [9]	1.14 MiB	0.021	90.7	100.0	100.0	100.0	100.0
RawTFNet [10]	694 KiB	0.046	71.8	100.0	100.0	100.0	100.0
W-MesoNet [27]	29.2 MiB	0.065	84.8	96.5	97.0	94.0	100.0
Proposed	151 KiB	0.021	98.4	89.0	99.5	82.5	90.0

Table 2. Leave-one-generator-out EER (%) for proposed detector.

Held out	CosyV.	Tort.	Fish	Parler	StyleT2	OpenV.	Melo
Test clips	8,309	10,388	2,853	1,944	2,024	7,211	10,388
EER (%)	0.014	0.288	1.046	3.245	3.845	5.065	6.747

We compare against four compact waveform detectors reported in [1]. All four use the same data splits and are trained for up to 100 epochs, with early stopping after 10 epochs without improvement in validation EER. Sampling assigns equal probability to human and synthetic speech and balances sources within each class.

Cross-Source Transfer. We evaluate all five detectors without retraining and keep their decision thresholds fixed before evaluating the unseen sources. Human speech consists of 1,000 English read-speech recordings from LibriSpeech [22]. Synthetic speech contains 200 recordings generated by each of four additional TTS systems listed in Table 1. The LibriSpeech recordings do not overlap with the LibriSeVoc recordings used for training or seen-source evaluation.

Leave-One-Generator-Out. To examine whether the proposed representation depends on generator-specific cues, we retrain the detector in seven folds, each excluding one SpeechFake BD generator from training. Human training data, feature extraction, and classifier hyperparameters remain unchanged. Each fold uses 1000 boosting iterations without validation-based checkpoint selection. We compute EER on the excluded generator together with the same 43,117 human test recordings for every fold (Table 2).

Preprocessing. All evaluations use 16-kHz mono audio and the leading 4-s segment of each recording. Only recordings long enough to provide a complete input segment are included. We apply no augmentation, loudness normalization, silence trimming, or additional lossy encoding.

4.2. Detection Performance and Generalization

Seen-Source Performance. The proposed detector achieves an EER of 0.021%, matching AASIST, the best-performing neural comparator in this evaluation, while requiring an estimated 151 KiB of tree storage compared with 694 KiB–29.2 MiB for the neural models (Table 1). This result shows that explicit acoustic measurements can provide highly discriminative evidence on seen speech sources without learning a waveform feature extractor end to end.

Cross-Source Transfer. Performance becomes more source dependent when the trained models are applied without retraining to unseen speech sources. The proposed detector correctly accepts 98.4% of the unseen LibriSpeech recordings as human, compared with 71.8–90.7% for the neural comparators. Its recall on unseen synthetic speech, however, varies from 82.5% on Qwen3-TTS to 99.5% on CosyVoice2, whereas the neural comparators reach 94–100% across the four synthetic sources. The resulting asymmetry is informative: the explicit representation produces relatively few false alarms on unseen human speech, but transfer to unseen generators is less uniform. Thus, generator shift, rather than seen-source discrimination, is the more important limitation of the current screen.

Table 3. Screening operating points selected from the held-out test ROC under the specified empirical FNR limits.

FNR limit (%)	0.000	0.010	0.020	0.050
FPR (%)	0.359	0.058	0.021	0.009

Held-Out Generator Generalization. The LOGO experiment further isolates this effect. EER ranges from 0.014% when CosyVoice is excluded from training to 6.747% when MeloTTS is excluded, with an average of 2.893% (Table 2). The wide range indicates that cues learned from the remaining generators transfer strongly to some held-out systems but not to others. This is consistent with the decoder analysis in Section 3.1.1: related synthesis operations can create shared spectral structure, while learned filters, intermediate representations, and waveform reconstruction can change the strength and shape of the artifacts. These results motivate evaluating the screen at explicit miss-rate constraints rather than relying on EER alone.

4.3. Screening Efficiency

Because the proposed model is designed as a front-end screen, its practical benefit depends on how many recordings can bypass expensive backend inference while maintaining a low synthetic-speech miss rate. We therefore evaluate a two-stage cascade in which our model screens every request and forwards suspected synthetic speech to *DF_Arena_IB_V_I* (DF), a 1.15-billion-parameter backend detector [1]. For a workload with synthetic-speech prevalence p , the fraction forwarded to DF is $q = (1 - p)\text{FPR} + p(1 - \text{FNR})$. Using a reference workload with $p = 0.1$, the 0.050% miss-rate operating point in Table 3 gives $q \approx 0.100035$, allowing about 90% of requests to bypass the backend.

We measure DF inference energy on an NVIDIA A100-SXM4-80GB using 256 randomly selected, class-balanced 4-s recordings at batch size 1. Across three post-warmup measurement windows, integrating GPU board power above the preceding idle baseline gives 10.422 J/request, including input transfer and score return. This subset is used only to estimate backend inference cost; forwarding rates are computed from the full test set. Front-end CPU costs are estimated from single-core AMD Milan runtimes assuming 15 W: audio decoding requires 0.310 J/request in both workflows, while feature extraction and tree classification add 0.324 J/request.

Let E_{decode} , E_{our} , and E_{DF} denote the corresponding per-request costs. Direct backend detection requires $E_{\text{direct}} = N(E_{\text{decode}} + E_{\text{DF}})$, whereas the two-stage cascade requires $E_{\text{screen}} = N(E_{\text{decode}} + E_{\text{our}} + qE_{\text{DF}})$. For $N = 10^6$ requests, estimated energy decreases from 2.981 to 0.466 kWh, an 84.4% reduction. The reduction primarily results from avoiding backend inference on benign requests while maintaining a stringent synthetic-speech miss rate. This estimate assumes constant DF energy per request and excludes model loading, warmup, GPU idle energy, and DF host-side costs.

5. CONCLUSION

In this paper, we presented a compact synthetic-speech screening approach that combines decoder-guided spectral artifacts with complementary spectro-temporal features. The resulting detector achieves 0.021% seen-source EER with an estimated model size of 151 KiB, while a two-stage screening analysis shows the potential to substantially reduce expensive backend inference. Results on unseen generators further highlight that cross-generator transfer remains the main challenge, motivating future work on more robust artifact representations and adaptive screening across evolving synthesis systems.

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